College of Business
香港城市大學
City University of Hong Kong

FinTech and AI in Finance

Class 6: FinTech Overview and Blockchain

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Semester A, 2026/27

City University of Hong Kong

FinTech Businesses

What Do We Need Financial Services For?

Consider a small retailer buying stock and selling online.

- **Receive payments.** Customers need a way to pay, and the shop needs access to the money.
- **Finance purchases.** Suppliers may need payment before the shop sells its stock.
- **Manage cash.** The owner keeps enough money for wages and decides what to do with spare funds.

Banks and other financial firms provide these services. Where could technology improve them?

Why Are Financial Services Costly?

The same retailer applies for a loan and accepts customer payments.

- **Information.** The lender checks who owns the business and whether sales can support repayment.
- **Processing.** Staff handle documents and payment problems, even for a small account.
- **Coordination.** The shop, payment provider, and bank compare records when amounts do not match.

These costs can make small transactions expensive to serve.

What Can Technology Change?

FinTech uses technology to improve how financial services are provided.

- **Data can support lending decisions.** With the retailer's permission, digital sales records help the lender assess cash flow.
- **Automation can reduce manual work.** Software matches incoming payments to orders and flags exceptions.
- **Connected systems can reduce repeated checks.** The shop and its providers exchange transaction updates.

Customers may gain quicker service or lower fees. The provider must cover the cost of the technology.

Digital payments

Class 10



Alipay

A mobile app for payments and other services.



Stripe

Payment processing for merchants.

Case focus

Who pays the fee, and how does the service compare with existing payment methods?

Lending and crowdfunding

Class 11



WeBank

A digital bank lending to small businesses.



Kickstarter

Funding creative projects in return for rewards.

Case focus

What does the funder receive, and who bears the risk of non-payment or project failure?

Robo-advice

Class 12

Betterment

Betterment

Automated investment portfolios in the US.

AQUMON

AQUMON

HKSAR digital advice. Brief comparison case.

Case focus

Can advisory fees cover the cost of serving small accounts?

Cybersecurity and RegTech

Class 12

Darktrace

DARKTRACE

Software to detect cyber threats.

RegTech applications: customer checks and financial-text review.

Case focus

What evidence shows lower losses or review costs, after accounting for false alerts?

Wallets and crypto lending

Classes 7–8

Ledger

Hardware wallets for protecting private keys.

Aave

A protocol for lending against crypto collateral.

Case focus

Who controls the assets, and who bears losses?

Stablecoins

Class 9

Circle

USDC: a token linked to the US dollar.

Tether

USDT: another dollar-linked token.

Case focus

How do issuers earn revenue, and can reserves meet redemptions?

The Layers of a FinTech Product



- **Customer**
- **App interface**
- **Financial service**
  Account, loan, investment
- **Infrastructure**
  Payments and data
- **Governance and risk**
  Rules and controls

Transaction Volume and Fee Revenue

Illustrative payment platform: 100,000 transactions per month, averaging \$40 each.

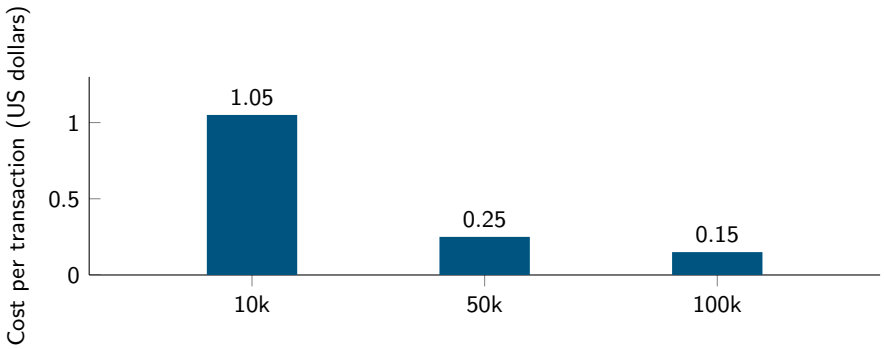
$$\text{Payment volume} = 100{,}000 \times \$40 = \$4 \text{ million}$$

$$\text{Revenue at a } 0.5\% \text{ fee} = \$4 \text{ million} \times 0.005 = \$20{,}000$$

- Payment volume measures money processed. Revenue is the fee the platform earns.
- Processing, customer support, fraud losses, and compliance must still be paid.

Scale and Average Processing Cost

Illustrative monthly costs: $10,000 fixed cost plus $0.05 per transaction.



$$\text{Average cost} = \frac{\text{Fixed cost}}{\text{Transactions}} + \text{Variable cost}$$

What Do Economists Mean by Money?

- **Medium of exchange:** people accept it as payment.
- **Unit of account:** prices and debts are stated in it.
- **Store of value:** it carries purchasing power into the future.

Which of these functions does Bitcoin perform well?

Bitcoin Began as a Two-Person Test



Satoshi Nakamoto 中本聡

- Published the design and released open-source software.
- Sent the first 10 BTC test transfer, then later stepped away.



Hal Finney

- Received the 10 BTC transfer on 12 January 2009.
- Independently ran the software, reported bugs, and helped improve it.

Governance: The network relies on shared rules and independent verification.

Sources: Nakamoto's Bitcoin white paper; Finney's account. AI-generated editorial portraits; Satoshi is an anonymous representation.

Bitcoin as Money

Function	Remarks
Medium of Exchange	**Partially**: Accepted in some online platforms and cross-border payments, but limited adoption in daily transactions.
Store of Value	**Contested**: Advocates call it "digital gold" due to scarcity (21M cap); critics emphasize extreme volatility.
Unit of Account	**Not yet**: Prices are almost always quoted in USD or other fiat currencies, not in BTC.

For our purposes: Bitcoin can perform some monetary functions, but not with the stability or broad acceptance of established money.

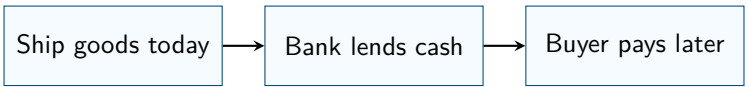
How a Blockchain Works

Ledgers, Blockchains, and Tokens

- A **ledger** records balances, ownership, or obligations. Banks and companies already keep ledgers.
- A **blockchain** links batches of records and uses agreed rules to maintain a shared history.
- A **token** is a digital unit recorded on that ledger. A shared invoice ledger need not have a publicly traded token.

The Exporter's Working-Capital Gap

Illustrative exporter: goods ship today, but the buyer pays in 60 days. This unpaid amount is a **receivable**.



- The exporter needs cash now to pay suppliers and workers.
- The bank verifies the receivable and decides how much to advance.
- Buyer payment repays the financing. A delivery dispute can still delay collection.

Two Banks, One Export Receivable

An exporter presents the same RMB 1m receivable to two banks.

Bank A's record

Already advanced RMB 600,000 against this receivable.

Bank B's decision

Receives a new request for RMB 600,000 against the same receivable.

Suppose total financing may not exceed 80% of the receivable. What does Bank B need to know before lending?

A shared record can reveal prior financing only if the participating banks report it.

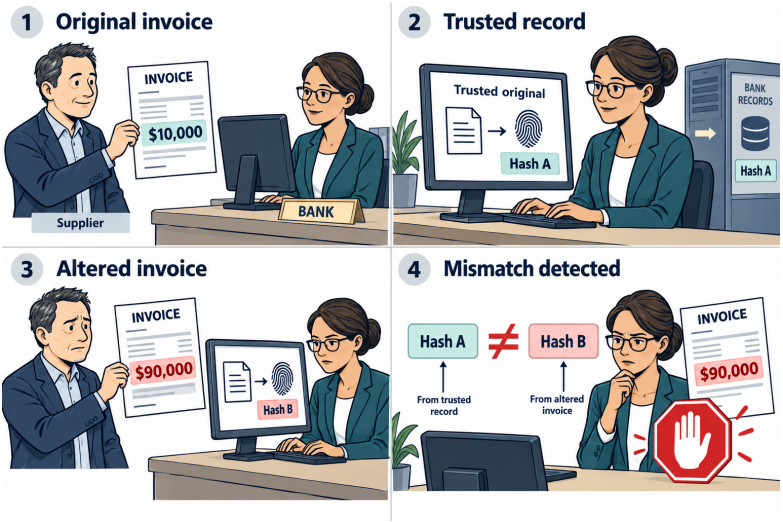
Hypothetical financing example. The 80% aggregate limit is a teaching assumption.

Who May Read and Update the Ledger?

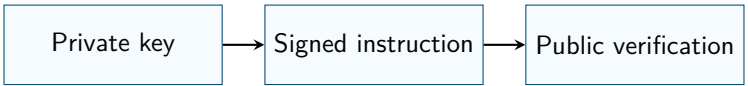
	Public network	**Permissioned network**
Participation	Open under protocol rules	Approved institutions
Records	Usually publicly inspectable	Access can be restricted
Governance	Protocol community	Named operators and members

Bitcoin is a public network. A bank consortium can restrict access to its trade records.

Hashing: A Digital Fingerprint of a Record



Private Keys, Public Keys, and Signatures



- A private key authorizes a transaction and stays with its holder.
- The public key allows others to verify the signature without learning the private key.
- A valid signature links the instruction to a key. It does not establish the holder's legal identity.

Source: Bitcoin.org, How Bitcoin works.

A Signature Exercise without Code

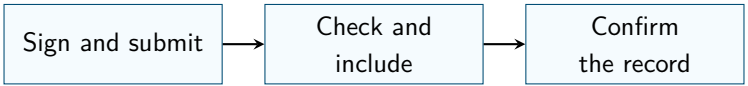
Message A
"Alice pays Bob 1 token."

Message B
"Alice pays Bob 10 tokens."

1. Alice signs the hash of Message A with her private key.
2. Bob verifies it with Alice's public key.
3. Changing 1 to 10 changes the hash, so the old signature fails.

The signature proves authorization and message integrity. It does not prove that Alice has enough funds.

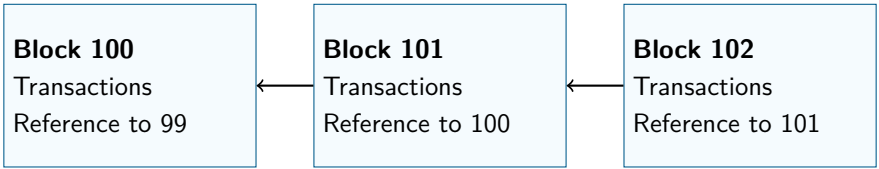
A Transfer from Instruction to Confirmation



- Alice's wallet submits an instruction to pay Bob.
- The network checks authorization and available funds before including the payment.
- Bob waits for the required confirmation before treating the payment as settled.

Source: Bitcoin.org.

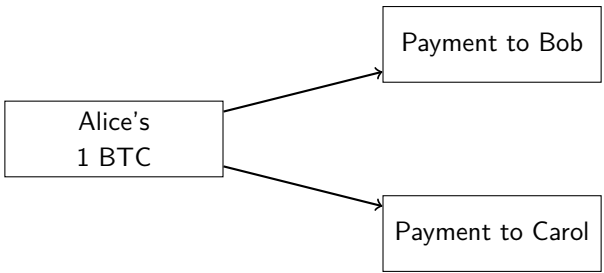
Blocks Link the Transaction History



- Each block contains a reference to the preceding block's hash.
- Changing an earlier record breaks the existing links. Consensus rules determine which valid history the network accepts.

The Double-Spending Problem

Example: Alice tries to spend the same 1 BTC twice.



Both instructions can carry valid signatures. The accepted ledger must allow at most one spend of the same funds.

Consensus and Economic Incentives

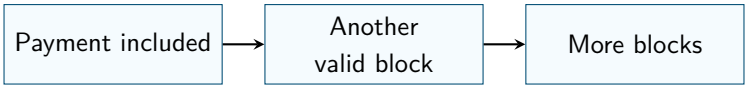
Consensus is how a network selects one valid transaction history.

	Bitcoin: proof of work	**Ethereum: proof of stake**
Resource committed	Computing equipment and electricity	ETH pledged as stake
Participants	Miners propose blocks	Validators propose and check blocks
Incentive	Rewards can cover mining costs	Rewards and penalties affect staked capital

Other nodes still check protocol rules. A block producer cannot make an invalid signature valid.

Source: Ethereum.org, Proof of stake versus proof of work.

Confirmations and Settlement Confidence



- In Bitcoin, later blocks make replacing an earlier accepted payment harder.
- A business may wait longer for a large payment than for a small one.
- Waiting reduces reorganization risk. It does not remove exchange-rate or custody risk.

Private Keys and Authorization Failures

Who can approve a transfer, and what exactly are they approving?

US$1.46bn

Bybit, February 2025

- A deceptive interface led staff to approve a change that gave attackers control of the wallet.
- **Control lesson:** independently check what an approval will authorize.

about US$620m

Ronin Bridge, March 2022

- Attackers controlled five of the nine authorizing keys, enough to approve withdrawals.
- **Control lesson:** keep approval credentials and control over them independent.

Sources: Bybit incident timeline; FBI, Bybit theft attribution; U.S. DOJ, Ronin hack description. Loss values are incident-date estimates.

Multiple-Choice Question: Custody and Access Control

A US asset manager requires two employees to approve a crypto transfer, but both approval accounts share one administrator password.

Which change most directly improves control over unauthorized transfers?

A) Increase the number of approvers using the same administrator account.

B) Separate approval credentials and restrict who can change access permissions.

C) Move the assets to a blockchain with lower transaction fees.

D) Produce a more detailed monthly report after transfers have occurred.

Recorded Information and Business Reality

The system may establish	The business must still establish
An invoice file has not changed	Goods actually shipped
An authorized key signed a payment	The employee had approval to pay
A loan already appears in the record	The borrower can repay it

When a Shared Ledger Is Useful

- Several independent organizations update or use the same records.
- Disagreements between their records create delays or costs.
- They can agree on access, update rules, and responsibility for corrections.

If one trusted organization controls the whole process, a conventional database may be simpler.

Multiple-Choice Question: Export Finance in China

A Chinese exporter submits an invoice recorded on a shared blockchain platform. A bank is considering a loan against the unpaid invoice.

Which check is most important before the bank lends?

A) Confirm only that the uploaded file has remained unchanged.

B) Use the platform's user count as evidence of the exporter's credit quality.

C) Accept the invoice because all participating banks use the same software.

D) Verify the trade documents and check financing already linked to the receivable.

Multiple-Choice Question: A Shared Ledger for US Banks

Several US banks consider a shared ledger to replace separate records of the same interbank payments. The payment contracts and funding arrangements stay unchanged.

Which benefit should they expect from consistent shared records?

A) Each bank needs less funding because payment obligations disappear.

B) Counterparty default losses become the platform operator's responsibility.

C) Staff spend less time comparing inconsistent records and resolving discrepancies.

D) Every customer payment becomes an insured bank deposit.

Blockchain in Financial Services

Blockchain Applications in China

- **Trade finance:** The State Administration of Foreign Exchange (SAFE) uses a shared platform for banks to verify export documents and check existing financing.
- **Tax invoices:** Shenzhen Tax Bureau and Tencent's 2018 pilot linked invoice issuance with reimbursement records, helping detect repeated claims.
- **Judicial records:** People's courts use blockchain records to verify electronic documents and evidence for changes after recording.

Sources: SAFE Tianjin, September 2025; Shenzhen Government, August 2021; Supreme People's Court, May 2022.

SAFE: Export Receivables Financing

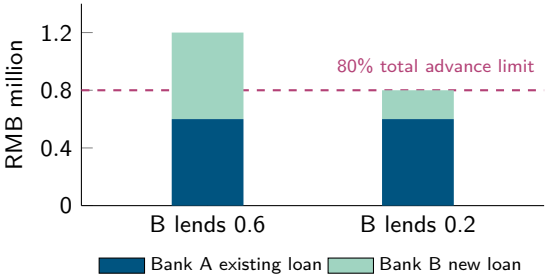
An exporter ships goods today but receives payment later. A bank can lend against this **receivable**, the money the overseas buyer owes.

1. **Financing application.** The exporter submits trade documents through SAFE's Cross-border Financial Service Platform.
2. **Bank verification.** The bank checks customs documents and prior financing. With authorization, it can also consult the borrower's foreign-exchange receipts and payments.
3. **Shared records.** Participating banks record lending and repayment. These records support the bank's own credit decision and checks for repeated financing.

Sources: SAFE Tianjin, September 2025, pp. 42 and 44; SAFE Hubei, April 2020.

Worked Example: Duplicate Receivables Financing

One receivable of **RMB 1 million**. Bank A has lent **RMB 600,000**. The exporter asks Bank B for another **RMB 600,000**.



Assumed total financing limit

$$80\% \times 1{,}000{,}000 = 800{,}000$$

Remaining capacity for Bank B

$$800{,}000 - 600{,}000 = 200{,}000$$

The shared record lets Bank B see Bank A's existing loan.

The bank still needs to assess repayment risk before lending.

Illustrative amounts and a shared 80% advance limit for this example, not a SAFE lending rule.

Worked Example: Financing a Receivable

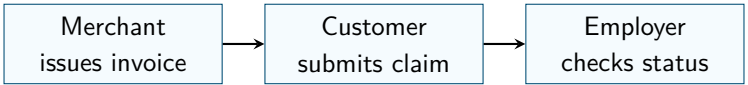
Illustrative terms: RMB 600,000 for 60 days at 6% per year, using a 360-day basis. No other fees.

$$\text{Interest} = 600{,}000 \times 6\% \times \frac{60}{360} = \boxed{\text{RMB } 6{,}000}$$

- Faster document verification can make the loan available sooner.
- The amount borrowed, interest rate, and financing period still determine the borrower's interest bill.

Shenzhen: Electronic Invoice Records

Documented application: Shenzhen's blockchain invoice programme linked issuance and reimbursement records.



- A common invoice identifier helps trace the same claim across connected systems.
- The business benefit is less repeated entry and checking, including detection of duplicate reimbursement.

Source: Shenzhen Government, August 2021; historical application.

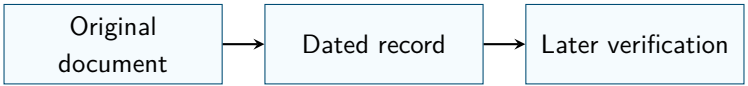
Worked Example: Two Claims for One Invoice

Illustrative expense claims

Claim	Invoice ID	Amount	Existing status
Employee A	SZ-1048	RMB 800	Reimbursed
Employee B	SZ-1048	RMB 800	Submitted again

- Matching the invoice ID flags the second claim for review.
- The system needs an up-to-date reimbursement status, not just an unchanged invoice file.
- Blocking the duplicate avoids an extra RMB 800 payment in this example.

China: Electronic Evidence in Courts

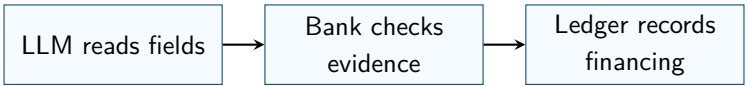


- Court blockchain systems support checks on stored evidence and case records.
- A record can help show that a file has not changed after registration.
- The court still assesses its source and relevance to the dispute.

Source: Supreme People's Court, May 2022.

LLMs and Shared Trade Records

Illustrative application to an export invoice



Extracted field	Example	Check
Invoice ID	EX-204	Match original document
Amount	RMB 1 million	Verify amount and currency
Payment due	60 days	Match sales contract

Extraction makes the document easier to process. The lender checks the evidence before recording the loan.

J.P. Morgan: Corporate Cash across Time Zones

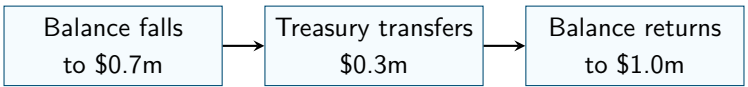
JPMorganChase

- Corporate treasurers need cash in the right subsidiary and currency. Banking cutoffs and manual transfers can leave cash idle elsewhere.
- Kinexys Digital Payments uses blockchain deposit accounts for near real-time transfers within its supported network.
- Siemens expanded into on-chain FX in October 2025. The service remains bank money, with account and currency restrictions.

Source: J.P. Morgan, Siemens and B2C2 mandates, October 2025.

J.P. Morgan: Programmable Treasury Payments

Illustrative treasury rule: maintain a $1 million minimum balance in a subsidiary's account.



- Kinexys supports programmable payments. A balance rule can replace repeated manual transfer instructions.
- The bank supplies the payment service. The corporate client sets permissions and supplies the cash.

Illustrative amounts, not Siemens account data. Source: J.P. Morgan, Kinexys programmable payments.

J.P. Morgan Case: The Value of Faster Cash Transfers

Illustrative Kinexys client decision: average idle cash falls from \$2.0m to \$1.2m.



$$\text{Annual opportunity-cost saving} = \$0.8m \times 4\% = \boxed{\$32{,}000}$$

Hypothetical saving, not a reported client result. At a 4% alternative return, annual fees and implementation costs must be below the benefit.

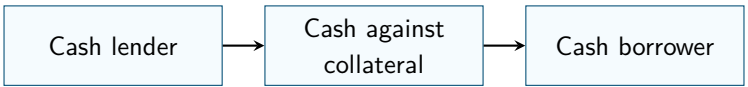
Short-Answer Question: Evaluating a Trade-Finance Platform

A bank in China uses a shared platform to verify export documents and check previous financing. Loan officers still decide whether to lend.

1. Explain how the platform could reduce duplicate financing.
2. Identify one reason a verified export receivable might still go unpaid.
3. Name one measure the bank could compare before and after adoption.

Broadridge: Funding a Dealer during the Day

Repo is short-term borrowing backed by securities, with an agreement to reverse the transaction later.

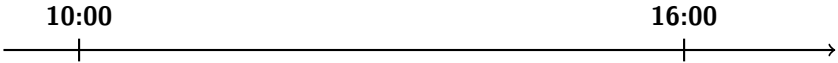


- Dealers may need cash for hours rather than overnight. Separate cash and securities records make short trades costly to coordinate.
- Broadridge's Distributed Ledger Repo (DLR) coordinates collateral and cash on a shared record, including same-day repayment.

Source: Broadridge, Intraday repo for liquidity providers.

Broadridge DLR: Settlement and Commercial Use

Illustrative transaction: a dealer needs $12 million between 10:00 and 16:00.



DLR reported **$368bn average daily repo volume in April 2026**. This measures transactions processed, not Broadridge revenue.

Source: Broadridge, 4 May 2026. Timeline amounts are illustrative.

Broadridge Case: Would a Bank Pay for DLR?

Hypothetical DLR adoption decision: $300,000 setup cost.



$$\text{Net annual saving} = \$180{,}000 - \$60{,}000 = \$120{,}000$$

$$\text{Simple payback} = 300{,}000/120{,}000 = \boxed{2.5 \text{ years}}$$

Teaching assumptions, not a Broadridge quote or client result. Integration costs and enough participating counterparties determine adoption. Excludes discounting and taxes.

TradeLens: Would Your Company Join?

Maersk and IBM launched TradeLens in 2018 to share shipping information across organizations.

- **Carrier:** better shipment visibility, but who can access commercially sensitive records?
- **Port or freight forwarder:** less repeated paperwork, but who pays for system integration?
- **Cargo owner:** more complete tracking, but which carriers and ports actually contribute data?

Choose one participant and state a condition for joining.

Source: Maersk, TradeLens announcement, 29 November 2022. Participant questions are classroom analysis.

TradeLens: Why the Network Did Not Continue

November 2022: Maersk and IBM announced that they would discontinue TradeLens.

- Maersk cited insufficient global industry collaboration and commercial viability.
- Each participant bears its own integration costs. The value of joining depends on participation elsewhere in the network.
- A conventional database may suffice when participants accept one operator. A shared ledger still needs agreed access, funding, and exit rules.

Source: Maersk, 29 November 2022. The final two points are business-model analysis, not a causal study of the closure.

Evaluating an Export-Finance Pilot

Illustrative results for comparable applications

Measure	Before	After
Manual handling time per application	40 minutes	25 minutes
Applications needing record reconciliation	12%	4%
One-year default rate	2.0%	2.1%

Processing improved in this example. The table does not establish lower credit risk or a causal effect without a comparable control group.

Worked Example: Verification Cost and Loan Profit

Illustrative one-year loan: RMB 100,000 at 8%. Funding costs RMB 4,000 and expected credit losses are RMB 2,000.

	Before	**After**
Interest income	8,000	8,000
Funding and expected credit losses	6,000	6,000
Verification and servicing cost	1,500	1,000
Expected profit (RMB)	**500**	**1,000**

Lower processing cost raises profit even if borrower default risk is unchanged.

The Same Technology Meets Two Policy Boundaries

Chinese Mainland

- supports business uses of blockchain and separate e-CNY initiatives;
- prohibits domestic virtual-currency exchange and ICO services. Bitcoin and Ether are not legal tender.

HKSAR

- licenses trading platforms and permits regulated access, including spot virtual-asset ETFs listed in 2024;
- develops tokenized deposits and bonds.

Sources: PBoC et al., Notice [2026] No. 42; SFC VATP framework; HKEX, 30 April 2024.